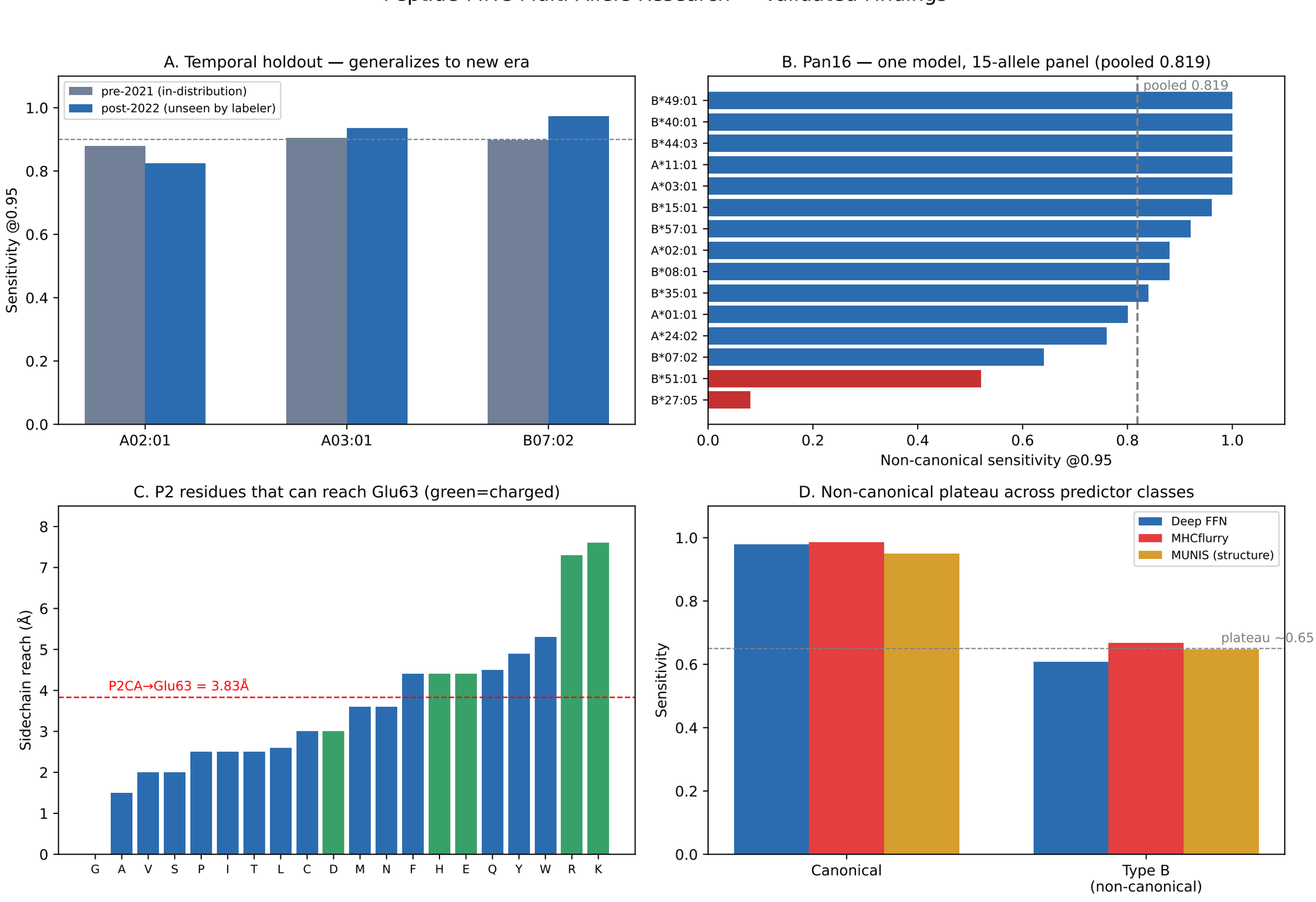
Peptide-MHC Multi-Allele Research — Validated Findings
A. Temporal holdout — generalizes to new era
pre-2021 (in-distribution)
post-2022 (unseen by labeler)
Sensitivity @0.95
A02:01
A03:01
B07:02
B. Pan16 — one model, 15-allele panel (pooled 0.819)
pooled 0.819
B*49:01
B*40:01
B*44:03
A*11:01
A*03:01
B*15:01
B*57:01
A*02:01
B*08:01
B*35:01
A*01:01
A*24:02
B*07:02
B*51:01
B*27:05
Non-canonical sensitivity @0.95
C. P2 residues that can reach Glu63 (green=charged)
Sidechain reach (Å)
P2CA→Glu63 = 3.83Å
G A V S P I T L C D M N F H E Q Y W R K
D. Non-canonical plateau across predictor classes
Deep FFN
MHCflurry
MUNIS (structure)
Sensitivity
plateau ~0.65
Canonical
Type B
(non-canonical)